

2-Mark Questions & Answers

Q1. Define Computer Architecture and Computer Organization.

* **Computer Architecture:** Refers to the functional attributes of a computer system as seen by the programmer (e.g., instruction set, data types, addressing modes).

* **C
specific

Q2. What is the Von Neumann bottleneck?

The Von Neumann Bottleneck is a speed limitation caused by sharing a single physical bus for both instruction fetch and data transfer. Since the CPU cannot fetch an instruction and read/write data simultaneously, the throughput of the system is constrained by the speed of the shared bus.

Q3. Define CPI and IC with formulas.

* **CPI (Cycles Per Instruction):** The average number of clock cycles required to execute a single instruction.

* `CPI

Q4. What is the stored program concept?

The Stored Program Concept (proposed by Von Neumann) states that both program instructions and data are stored together in the same main memory (RAM) in binary format. This allows the computer to fetch, decode, and execute instructions sequentially without requiring physical hardware rewiring.

Q5. State two differences between Von Neumann and Harvard architecture.

| Feature | Von Neumann Architecture | Harvard Architecture |

| :--- | :-

Q6. What are the two units of the 8086 microprocessor?

1. Bus Interface Unit (BIU): Handles all bus operations, address calculation, memory fetching, and instruction queuing.

2. Exec

Q7. Define MIPS and MFLOPS.

* **MIPS (Millions of Instructions Per Second):** A measure of processor execution speed.

* `MIP

Q8. What is the size of the pre-fetch queue in 8086?

The 8086 microprocessor features a 6-byte pre-fetch instruction queue located inside the Bus Interface Unit (BIU). It prefetches up to 6 bytes of instruction code from memory to enable instruction pipelining.

Q9. State the syntax and one example of immediate addressing mode.

In Immediate Addressing Mode, the operand data is specified directly within the instruction itself.

* **Syntax

Q10. Differentiate between minimum mode and maximum mode of 8086.

* **Minimum Mode (MN/MX' = 5V):** Configured for single-processor systems; the 8086 directly generates all bus control signals itself.

* **Maximum Mode
signals

Q11. What is the function of BIU in 8086?

The Bus Interface Unit (BIU) manages all external data, address, and control bus operations. Its primary functions are:

1. Fetch

Q12. Define latency and throughput.

* **Latency (Execution Time):** The total time required to complete a single task from start to finish (measured in seconds or milliseconds).

* **Throughput
